Goru Shanmukha

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Professional Summary

Computer Science undergraduate at IIIT Kancheepuram with applied experience in machine learning fundamentals — building and evaluating supervised and unsupervised learning models from scratch. Proficient in model training, evaluation, and data preprocessing pipelines, with a working knowledge of probability, statistics, and linear algebra as applied to ML systems. Focused on developing practical problem-solving skills across deep learning and data-driven analysis.

Education

Indian Institute of Information Technology Design and Manufacturing 2023 – 2027
B.Tech in Computer Science and Engineering · CGPA: 7.89 *Kancheepuram, India*

Skills

Languages Python, C++, SQL
ML & AI Supervised Learning, Unsupervised Learning, CNNs, K-Means Clustering, Classification, Deep Learning, Model Training & Evaluation
Mathematics Probability, Statistics, Linear Algebra, Feature Engineering
Libraries PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Streamlit
Core CS Data Structures & Algorithms, OOP, DBMS
Dev Tools Git, GitHub, VS Code, Google Colab

Projects

Facial Emotion Recognition System | *Python, PyTorch, CNN, Deep Learning* 2026

- Designed and trained a **supervised learning** CNN using **cross-entropy loss** and the **Adam optimizer** on **35,887 labeled grayscale images** from FER2013, achieving **~65% classification accuracy** across 7 emotion classes.
- Designed **data preprocessing** and augmentation pipelines (normalization, random flipping, cropping) in Python/PyTorch as a **regularization** strategy, reducing **overfitting** and improving model generalization across training epochs.
- Built **model evaluation** workflows using Matplotlib and Pandas to track per-class performance, visualize confusion matrices, and identify failure modes in ambiguous expressions.
- Exported trained model checkpoints for reusable inference, enabling reproducible experimentation and **model deployment** without full retraining.

Country Segmentation for Development Aid Allocation | *Python, Scikit-learn, K-Means* 2025

- Applied an **unsupervised learning** K-Means model to **167 countries** across socio-economic indicators (GDP/capita, child mortality, income), surfacing **3 distinct development tiers** for targeted aid analysis.
- Conducted **exploratory data analysis (EDA)** and **feature engineering** — scaling, correlation filtering, and elbow-curve evaluation — to tune the clustering model and validate segment separation quality.
- Produced **data visualizations** using Matplotlib and Seaborn to communicate cluster boundaries and flag the highest-need segment for priority aid consideration.
- Delivered actionable findings: the lowest-income cluster — dominated by Sub-Saharan African nations — showed the strongest correlation between child mortality and GDP deficit, supporting targeted intervention recommendations.

Achievements

Solved **300+ algorithmic problem-solving** challenges on LeetCode and HackerRank (graphs, dynamic programming, trees); supplemented with self-study in **probability, statistics, and linear algebra** to strengthen mathematical foundations for ML model development.